Advanced Programming Practice · 21CSC203P

Unit 5 · Session 3 · SLO 2 — Solved Worksheet

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Unit â€“ 5 : Session â€“ 3 : SLO â€“ 2 | Simulating Pattern Recognition with Input Strings

I. Questions & Answers

1. DFA for "ends in 01" â€” accepts exactly the strings over {0,1} whose last two symbols are 01, e.g. 01, 001, 101, 1101, 10101. It rejects Îµ, 0, 1, 00, 10, 11, 100, etc.

2. Simulating input '1010' (DFA ends-in-01, states q0=start/no-match, q1=last symbol 0, q2=accept/ends-01): read 1 â†’ q0; read 0 â†’ q1; read 1 â†’ q2 (accept state); read 0 â†’ q1. String exhausted at q1, which is not accepting â†’ '1010' is REJECTED (it ends in 10, not 01).

3. Testing pattern recognition in code â€” with automata-lib, run dfa.read_input_stepwise("1010") to trace the state at each symbol, or dfa.accepts_input("1010") for the verdict; a hand-written simulator loops: state = Î´(state, symbol) for each character, then checks state â^ F.